CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy

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Author contribution statement

G.S. conceived the dataset and was its intellectual lead. He designed the label scheme, implemented the processing pipeline and the quality-control procedures, and wrote the manuscript. A.S. led the data annotation, annotated data, and edited the manuscript. A.T., M.K., R.C., D.H., F.M., H.S., M.S., S.S., M.M. and J.K. annotated data. M.I. and N.A. performed the Castellvi grading, each reading every case independently and resolving disagreements by consensus, and trained the student annotators in CT reading. G.S. reviewed the candidate instrumentation. Every author therefore contributed to the acquisition of the data. All authors revised the manuscript critically for important intellectual content, approved the final version, and agree to be accountable for all aspects of the work, so that questions about the accuracy or integrity of any part of it are investigated and resolved. The corresponding author, a faculty surgeon, holds supervisory responsibility for the student and resident co-authors.

Conflict of interest statement

The authors have no relevant conflicts of interest to disclose.

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Data and code availability (unblinded)

Dataset, archived before submission: https://doi.org/10.5281/zenodo.22139642 (concept identifier, resolves to the current version); the version described in the manuscript is v10, https://doi.org/10.5281/zenodo.22642578.
Code: https://github.com/Gregory-Schwing-MD-PhD/CTSpinoPelvic1K, tagged at the release commit.

Mirror: `https://huggingface.co/datasets/OpenSpineConsortium/CTSpinoPelvic1K`, a convenience mirror, not the archive of record.

Pseudolabel model checkpoints (Sec. II.F of the manuscript): `https://huggingface.co/OpenSpineConsortium/spinopelvic-seg-checkpoints`. The five folds were trained with `https://github.com/Gregory-Schwing-MD-PhD/spinesurg-ct-nnunet` at commit `50b209f` on `master` (2 May 2026; training began 3 May 2026), trainer `nnUNetTrainerWandB_500ep_LSTVOversample`, plans `nnUNetResEncUNetPlans_100G`, configuration `3d_fullres`, 500 epochs per fold, on the nine-class merged dataset (Dataset803); inference used `checkpoint_best.pth` of the fold that held each record out. The inference code and a released copy of the trainer are at `https://github.com/OpenSpineConsortium/spinopelvic-seg`.

Measurement toolkit, which computes every per-level dimension reported in the manuscript: `https://github.com/OpenSpineConsortium/OpenSpineToolkit`.

Build archive, every input to v10 that no script regenerates (source-mask placements, the model-output base the students corrected, the review ledgers, the hand corrections and the hardware reads), with the step-by-step chain from the public sources: `https://doi.org/10.5281/zenodo.22647933`.

Ethics

This work uses publicly available, de-identified imaging and annotations derived from it. The Wayne State University Institutional Review Board issued a Notice of IRB Administrative Determination of Non-Human Participant Research on 2 September 2026 (WSU IRB HPR Number 2026-269, “Open Spine Consortium: Secondary Analysis of Publicly Available, De-Identified Imaging Datasets and the Derived Annotations, Software, and Models Produced From Them”), finding that the project does not constitute human participant research under the Common Rule at 45 CFR 46 and that IRB review and oversight are not required.